

Yu-Ji Kim

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[Website](#) | [Scholar](#) | [GitHub](#) | [LinkedIn](#)

Research Interests

Embodied reasoning & navigation, 3D scene understanding, 3D reconstruction, Multi-modal understanding.

Experience

NVIDIA Research

Research Intern

Taipei, Taiwan

Feb. 2026 – Present

POSTECH → KAIST

Research Assistant (Advisor: Prof. Tae-Hyun Oh)

Pohang → Daejeon, Korea

Sep. 2021 – Present

NAVER Corporation

Backend Development Intern

Gyeonggi, Korea

Apr. 2020 – Jul. 2020

World Friends ICT Volunteers

Teaching Volunteer

Can Tho, Vietnam

Jun. 2018 – Aug. 2018

Education

Pohang University of Science and Technology, POSTECH

Integrated M.S.&Ph.D. in Graduate School of Artificial Intelligence (GSAI)

Pohang, Korea

Sep. 2021 – Present

Daegu Gyeongbuk Institute of Science and Technology, DGIST

Bachelor of Engineering (Track | Computer Science)

Daegu, Korea

Mar. 2016 – Aug. 2021

UC Berkeley, California

Summer Session

California, USA

Jun. 2017 – Aug. 2017

Publications (International)

“ * ” denotes equal contribution. “ † ” denotes co-corresponding authors.

[C5] Vision-Language Navigation with Action Grounded 3D Cognitive Maps, **Under Review, 2026.**

Kim Yu-Ji, TBA, J. Choe

(Work done during an internship at NVIDIA Research.)

[C4] SplatReasoner: Enhancing Embodied Reasoning and Grounding by Novel View Synthesis, **ECCV, 2026.**

Kim Yu-Ji, D. Lee, K. Jun-Seong, N. Hyeon-Woo, G. Kim, Y. Kwon, Y.-C. F. Wang, J. Choe, T.-H. Oh

[C3] Dr. Splat: Directly Referring 3D Gaussian Splatting via Direct Language Embedding Registration, **CVPR, 2025.**

K. Jun-Seong, G. Kim, **Kim Yu-Ji**, Y.-C. F. Wang, J. Choe[†], T.-H. Oh[†]

(Accepted as a highlight paper (< 3% acceptance rate), Winner of Qualcomm Innovation Fellowship 2025 Korea)

[C2] MeTTA: Single-View to 3D Textured Mesh Reconstruction with Test-Time Adaptation, **BMVC, 2024.**

Kim Yu-Ji, H. Ha, K. Youwang, J. Surh, H. Ha[†], T.-H. Oh[†]

(Best Poster Award from BMVC 2024)

[C1] Scratching Visual Transformer’s Back with Uniform Attention, **ICCV, 2023.**

N. Hyeon-Woo, **Kim Yu-Ji**, B. Heo, D. Han, S.-J. Oh, T.-H. Oh

[C0] HDR-Plenoxels: Self-Calibrating High Dynamic Range Radiance Fields, **ECCV, 2022.**

K. Jun-Seong*, **Kim Yu-Ji***, M. Ye-Bin, T.-H. Oh

Award

Outstanding Student Paper Award from Autumn IEIE 2025 <i>“Text-Embedded 3DGS as Enhanced Language Embeddings.”</i>	Dec. 2025
Outstanding Reviewer from ICCV 2025 <i>High-quality reviews (as judged by their Area Chairs) placed them among the top 3% of reviewers.</i>	Oct. 2025
Best Poster Award from BMVC 2024 <i>“MeTTA: Single-View to 3D Textured Mesh Reconstruction with Test-Time Adaptation.”</i>	Nov. 2024

Academic Service

Conference Reviewer

- European Conference on Computer Vision (ECCV), 2026
- International Conference on Computer Vision (ICCV), 2025 (**Outstanding Reviewer 2025 (Top 3%)**)
- Annual Conference on Neural Information Processing Systems (NeurIPS), 2026
- International Conference on Learning Representations (ICLR), 2025
- AAAI Conference on Artificial Intelligence (AAAI), 2025
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024, 2026
- British Machine Vision Conference (BMVC), 2024
- IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026

Journal Reviewer

- The Visual Computer (TVCJ), 2022

Invited Talk

HDR-Plenoxels: High Dynamic Range Radiance Fields	Mar. 2023
• Workshop on Google ExploreCSR, POSTECH, Korea.	

Patent

Vision-Language Navigation Using 3D Cognitive Maps	2026
• App.; US 64/089,945	
Method and apparatus for generating 3 dimension high dynamic range radiance field	2023
• App.; US 18/503,784, Patent granted; KR 10-2742898	

Projects

3D Structured Memory with VLMs	Mar. 2025 – Dec. 2025
• 3D structured memory with VLMs for an embodied reasoning.	
• Supported by ETRI, South Korea.	
Spatial Data Exploration with Common Latent Features	May 2024 – Dec. 2024
• Joint RGB-Depth amodal completion with generative models.	
• Supported by ETRI, South Korea.	
Single Image to 3D Reconstruction	Jan. 2023 – Dec. 2023
• Developing a single image to 3D textured mesh reconstruction multi-view generative models.	
• Supported by Bucketplace, South Korea.	
Domain Adaptation for Object Detection	Nov. 2021 – Nov. 2022
• Developing a domain adaptation for object abnormal detection.	
• Supported by LG Display, South Korea.	

Teaching Experiences

NAVER Boostcamp AI Tech (4th, 5th, 6th, 7th, 8th)

2022 - 2025

- Participated as a production assistant for computer vision lectures.

[CSED232] Object-oriented Programming, POSTECH

Spring 2024

- Working as a teaching assistant for Prof. Sunghyun Cho.

[EECE695] Human-centric Social AI System Design, POSTECH

Fall 2022, Fall 2023

- Working as a teaching assistant for Prof. Tae-Hyun Oh.

Extra

Deep Learning Seminar

- KAIST AMI Lab Seminar (Seminar videos on [YouTube](#))

Reference

Dr. Prof. Tae-Hyun Oh, Professor, KAIST

- Relationship: Ph.D. advisor